

Technical Implementation Report:

Mammonto Building Projects Network

Infrastructure Deployment

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Project: Mammonto Building Projects Enterprise Network
Environment: Cisco Packet Tracer Architecture

1. Executive Overview

This document details the complete design, configuration, and verification of the enterprise network deployment for Mammonto Building Projects. The objective was to build a segmented, resilient, and scalable network topology capable of hosting corporate departments, mobile staff, and centralized infrastructure.

The core design implements 802.1Q VLAN segmentation, Router-on-a-Stick (RoaS) Inter-VLAN routing using a Cisco ISR4331 router, automated dynamic IP address assignment via Cisco IOS DHCP services, and secure WPA2-PSK/AES wireless integration for staff mobility.

2. IP Addressing & Topology Architecture

Subnet Scheme (10.32.0.0/16)

The network uses a structured 10.32.X.0/24 CIDR block mapped to functional departments:

VLAN ID	Subnet Name	Subnet Network	Subnet Mask	Defiault Gateway	Dynamic Host Range
VLAN 10	ADMIN	10.32.10.0/24	255.255.255.0	10.32.10.1	10.32.10.11 – 10.32.10.254
VLAN 20	PROJECT	10.32.20.0/24	255.255.255.0	10.32.20.1	10.32.20.11 – 10.32.20.254
VLAN 30	FINANCE	10.32.30.0/24	255.255.255.0	10.32.30.1	10.32.30.11 –

					10.32.30.254
VLAN 40	STAFF-WIFI	10.32.40.0/24	255.255.255.0	10.32.40.1	10.32.40.11 – 10.32.40.254
VLAN 99	MGMT	10.32.99.0/24	255.255.255.0	10.32.99.1	10.32.99.2 – 10.32.99.254

3. Step-by-Step Device Configurations

Phase 1: Core Router Configuration (Router-ISR4331)

Configured the primary interface for 802.1Q subinterfaces to act as default gateways across all segmented subnets, and established dedicated DHCP pools with reserved gateway blocks.

enable

configure terminal

! Bring up physical interface

interface GigabitEthernet0/0/0

no shutdown

exit

! Subinterface Setup for Inter-VLAN Routing

interface GigabitEthernet0/0/0.10

encapsulation dot1Q 10

ip address 10.32.10.1 255.255.255.0

exit

interface GigabitEthernet0/0/0.20

encapsulation dot1Q 20

ip address 10.32.20.1 255.255.255.0

exit

interface GigabitEthernet0/0/0.30

encapsulation dot1Q 30

ip address 10.32.30.1 255.255.255.0

exit

```
interface GigabitEthernet0/0/0.40
 encapsulation dot1Q 40
 ip address 10.32.40.1 255.255.255.0
 exit
```

```
interface GigabitEthernet0/0/0.99
 encapsulation dot1Q 99
 ip address 10.32.99.1 255.255.255.0
 exit
```

! DHCP Scope & Excluded Ranges Configuration

```
ip dhcp excluded-address 10.32.10.1 10.32.10.10
ip dhcp excluded-address 10.32.20.1 10.32.20.10
ip dhcp excluded-address 10.32.30.1 10.32.30.10
ip dhcp excluded-address 10.32.40.1 10.32.40.10
```

```
ip dhcp pool ADMIN_POOL
 network 10.32.10.0 255.255.255.0
 default-router 10.32.10.1
 dns-server 8.8.8.8
 exit
```

```
ip dhcp pool PROJECT_POOL
 network 10.32.20.0 255.255.255.0
 default-router 10.32.20.1
 dns-server 8.8.8.8
 exit
```

```
ip dhcp pool FINANCE_POOL
 network 10.32.30.0 255.255.255.0
 default-router 10.32.30.1
 dns-server 8.8.8.8
 exit
```

```
ip dhcp pool WIFI_POOL
 network 10.32.40.0 255.255.255.0
 default-router 10.32.40.1
 dns-server 8.8.8.8
 exit
```

```
end
write memory
```

Phase 2: Core Switch Configuration (Core-Switch-01)

Defined global VLANs, configured the out-of-band management interface, and provisioned 802.1Q trunk lines to handle tagged traffic between the router and access layer.

enable

configure terminal

! Database VLAN definitions

vlan 10

name ADMIN

vlan 20

name PROJECT

vlan 30

name FINANCE

vlan 40

name STAFF-WIFI

vlan 99

name MANAGEMENT

exit

! In-Band Switch Management Interface

interface vlan 99

ip address 10.32.99.2 255.255.255.0

no shutdown

exit

! Trunk Uplink to Core Router

interface gigabitEthernet0/1

switchport mode trunk

switchport trunk allowed vlan 10,20,30,40,99

no shutdown

exit

! Trunk Downlink to Switch-Access

interface gigabitEthernet1/1

switchport mode trunk

switchport trunk allowed vlan 10,20,30,40,99

no shutdown

exit

end

write memory

Phase 3: Access Switch Configuration (Switch-Access)

Created identical global VLAN definitions on Switch-Access and mapped physical edge ports directly to assigned host VLANs, applying spanning-tree portfast to bypass standard listening/learning states for end devices.

```
enable
configure terminal
```

! Global VLAN Creation

```
vlan 10
  name ADMIN
vlan 20
  name PROJECT
vlan 30
  name FINANCE
vlan 40
  name STAFF-WIFI
vlan 99
  name MANAGEMENT
exit
```

! Trunk Connection to Core Switch

```
interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
no shutdown
exit
```

! Port Configuration for Admin Devices (VLAN 10)

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
spanning-tree portfast
no shutdown
exit
```

```
interface fastEthernet0/2
switchport mode access
switchport access vlan 10
spanning-tree portfast
no shutdown
exit
```

! Port Configuration for Project Devices (VLAN 20)

```
interface fastEthernet0/3
switchport mode access
switchport access vlan 20
spanning-tree portfast
no shutdown
exit
```

! Port Configuration for Finance Devices (VLAN 30)

```
interface fastEthernet0/4
switchport mode access
switchport access vlan 30
spanning-tree portfast
no shutdown
exit
```

! Port Configuration for Wireless Access Point (VLAN 40)

```
interface fastEthernet0/6
switchport mode access
switchport access vlan 40
spanning-tree portfast
no shutdown
exit
```

end

write memory

Phase 4: Wireless LAN & Endpoint Provisioning

1. **Access Point Setup (AP01):** Connected to FastEthernet0/6 on Switch-Access (VLAN 40).
 - **SSID:** Staff-WiFi
 - **Authentication:** WPA2-PSK
 - **Encryption:** AES
 - **Passphrase:** Mammonto2026
2. **Mobile Device Provisioning:**
 - Configured **Laptop**, **Tablet**, and **Smartphone** clients with the Staff-WiFi profile parameters.
 - Interface configurations set to DHCP across all devices.

4. Verification & Testing Log

1. Dynamic Host Configuration Protocol (DHCP) Verification

Toggled IP configurations across wired and wireless nodes to verify lease generation from the ISR4331 router:

- **PC-ADMIN-01** (FA0/1): Received 10.32.10.21 /24, Default Gateway 10.32.10.1
- **PC-ADMIN-02** (FA0/2): Received 10.32.10.22 /24, Default Gateway 10.32.10.1
- **PC-PROJECT-01** (FA0/3): Received 10.32.20.21 /24, Default Gateway 10.32.20.1
- **PC-FINANCE-01** (FA0/4): Received 10.32.30.21 /24, Default Gateway 10.32.30.1
- **WiFi-LAPTOP-01 / Tablet / Smartphone**: Received 10.32.40.X /24 addresses, Default Gateway 10.32.40.1

2. Full Inter-VLAN Routing Verification

Executed command-line ICMP reachability checks from the wireless laptop to endpoints across all other corporate subnets:

Command Output: Inter-VLAN Reachability Test

Cisco Packet Tracer PC Command Line 1.0

```
C:\>ping 10.32.10.21
```

Pinging 10.32.10.21 with 32 bytes of data:

Reply from 10.32.10.21: bytes=32 time=1ms TTL=128

Reply from 10.32.10.21: bytes=32 time<1ms TTL=128

Reply from 10.32.10.21: bytes=32 time<1ms TTL=128

Reply from 10.32.10.21: bytes=32 time<1ms TTL=128

Ping statistics for 10.32.10.21:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

```
C:\>ping 10.32.20.21
```

Pinging 10.32.20.21 with 32 bytes of data:

Reply from 10.32.20.21: bytes=32 time=1ms TTL=128

Reply from 10.32.20.21: bytes=32 time<1ms TTL=128

Reply from 10.32.20.21: bytes=32 time<1ms TTL=128

Reply from 10.32.20.21: bytes=32 time<1ms TTL=128

Ping statistics for 10.32.20.21:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

```
C:\>ping 10.32.30.21
```

Pinging 10.32.30.21 with 32 bytes of data:

Reply from 10.32.30.21: bytes=32 time=1ms TTL=128

Reply from 10.32.30.21: bytes=32 time<1ms TTL=128

Reply from 10.32.30.21: bytes=32 time<1ms TTL=128

Reply from 10.32.30.21: bytes=32 time<1ms TTL=128

Ping statistics for 10.32.30.21:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.32.40.1

Pinging 10.32.40.1 with 32 bytes of data:

Reply from 10.32.40.1: bytes=32 time=1ms TTL=128

Reply from 10.32.40.1: bytes=32 time<1ms TTL=128

Reply from 10.32.40.1: bytes=32 time<1ms TTL=128

Reply from 10.32.40.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.32.40.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

5. Conclusion

The Mamkonto Building Projects network infrastructure is fully deployed, validated, and operational. All departmental VLANs are securely isolated at Layer 2 while maintaining full, routed Layer 3 connectivity across subnets. Wireless authentication and dynamic addressing function seamlessly across all wireless devices.